

1.4 States of Matter

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.4 States of Matter
Difficulty	Medium

Time allowed: 20

Score: /12

Percentage: /100

Question 1

An experiment is conducted to calculate the M_r of an unknown gas of known mass.

Measurements of pressure, volume and temperature are taken during the experiment.

Which conditions of temperature and pressure would give the most accurate value of M_r ?

	Pressure	Temperature
A	Low	High
B	Low	Low
C	High	High
D	High	Low

[1 mark]

Question 2

Three substances, X, Y and Z, have physical properties as shown.

Substance	Melting / °C	Boiling / °C	Electrical conductivity	
			of solid	of liquid
X	801	1413	Poor	Good
Y	2852	3600	Poor	Good
Z	3550	4827	Good	Not known

What could be the identities of X, Y and Z?

	X	Y	Z
A	CaF	KCl	Cu
B	NaCl	MgO	C (graphite)
C	NaBr	CaO	C (diamond)
D	NaBr	BaO	SiO ₂

[1 mark]

Question 3

0.96 g of hydrogen gas is contained in a sealed vessel of volume of $7.0 \times 10^{-3} \text{ m}^3$ at a temperature of 303 K.

Assume the gas behaves as an ideal gas.

What is the pressure of the vessel?

- A. 172.7 Pa
- B. 172.7 kPa
- C. 345.5 Pa
- D. 691.1 kPa

[1 mark]

Question 4

A 5370 cm^3 sample of oxygen is measured at a temperature of 60°C . The pressure measured was 103000 Pa.

Assume the gas behaves as an ideal gas.

What is the mass of the sample of oxygen?

- A. 3.2 g
- B. 6.4 g
- C. 1.8 g
- D. 3.6 g

[1 mark]

Question 5

A sample of chlorine gas with a mass 5.35 g has a volume of $1.247 \times 10^{-3} \text{ m}^3$ at a pressure of $1.00 \times 10^5 \text{ Pa}$.

Assuming the gas acts as an ideal gas, what is the temperature of the gas?

- A. 300 K
- B. 250 K
- C. 100 K
- D. 200 K

[1 mark]

Question 6

A bubble travelled up from the sea bed to the surface. Just below the surface of the sea the bubble had a volume of 200 cm^3 at a pressure of 101 kPa , the temperature at the surface is the same as the sea bed.

The pressure at the sea bed is 2020 kPa .

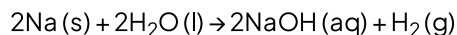
What is the volume of the bubble at the sea bed?

- A. 10 cm^3
- B. 200 cm^3
- C. 100 cm^3
- D. 1000 cm^3

[1 mark]

Question 7

Sodium reacts with water in the equation below.



Which mass of sodium reacts with water to produce 960 cm^3 of hydrogen gas under standard pressure and room temperature?

- A. 3.68 g
- B. 1.81 g
- C. 0.91 g
- D. 0.46 g

[1 mark]

Question 8

At 298 K and 101 kPa which compound is the only gas?

- A. Chloroethane
- B. 1-aminopropane
- C. Propan-1-ol
- D. Ethane-1, 2-diol

[1 mark]

Question 9

When a sample of a gas is compressed at room temperature from 101 kPa to 300 kPa, its volume changes from 50.0 cm³ to 17.5 cm³.

Which statement best describes why this happens?

- A. The gas behaves ideally.
- B. The gas behaves non-ideally.
- C. Gas is absorbed on to the vessel walls.
- D. The gas partially liquifies.

[1 mark]

Question 10

What mass of a gas would occupy a volume of 3 dm³ at 25 °C and 1 atmosphere pressure?

1 mol of a gas occupies 24 dm³ at 25 °C and 1 atmosphere pressure.

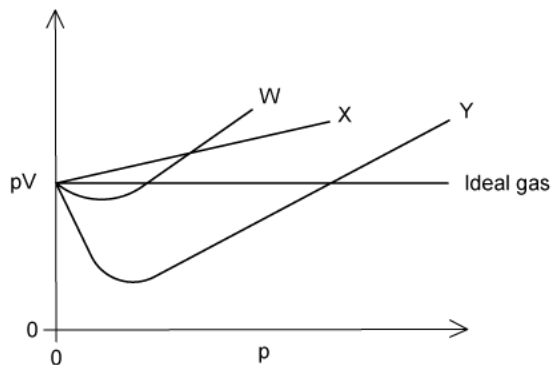
- A. 7.0 g of N₂
- B. 4.4 g of CO₂
- C. 4.0 g of He
- D. 2.0 g of CH₄

[1 mark]

Question 11

Real gases deviate from ideal behaviour. The extent to which they deviate depends on the identity of the gas.

Plots of pV against p for three real gases **W**, **X**, **Y** and an ideal gas are shown below.



What are the identities of gases **W**, **X**, **Y**?

	W	X	Y
A	N ₂	NH ₃	H ₂
B	N ₂	H ₂	NH ₃
C	H ₂	N ₂	NH ₃
D	H ₂	NH ₃	N ₂

[1 mark]

Question 12

Which expression gives the pressure exerted by 1.5×10^{-2} mol of carbon dioxide in a container of volume 2 dm^{-3} at 275°C ?

- A. $\frac{(1.5 \times 10^{-2}) \times 8.31 \times 275}{2 \times 10^{-6}}$
- B. $\frac{(1.5 \times 10^{-2}) \times 8.31 \times (275 + 273)}{2 \times 10^{-6}}$
- C. $\frac{(1.5 \times 10^{-2}) \times 8.31 \times (275 + 273)}{2 \times 10^{-3}}$
- D. $\frac{(1.5 \times 10^{-2}) \times 8.31 \times 275}{2 \times 10^{-3}}$

[1 mark]

Model Answers: Medium

1

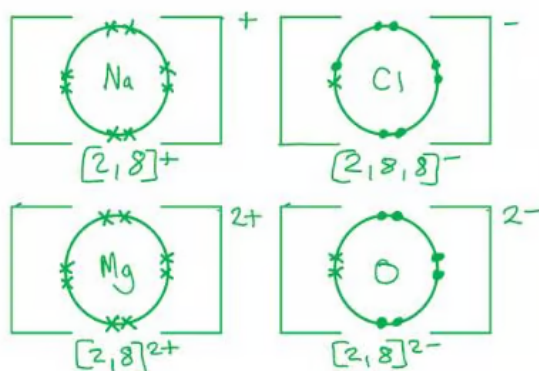
The correct answer is **A** because:

- The ideal gas equation is $pV = nRT$.
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).
- The closer the conditions are for ideal behaviour, the more accurate calculations are.
- Ideal behaviour is identical particles with no internal structure, no volume, random motion, no intermolecular forces and no loss of momentum in collisions.
- At low pressure, a gas minimises contact with other particles (intermolecular force tends towards zero as distance tends towards infinity).
- At high temperatures intermolecular forces are massively outweighed by its kinetic energy and any collision between particles is less likely to be inelastic (i.e. lose momentum).

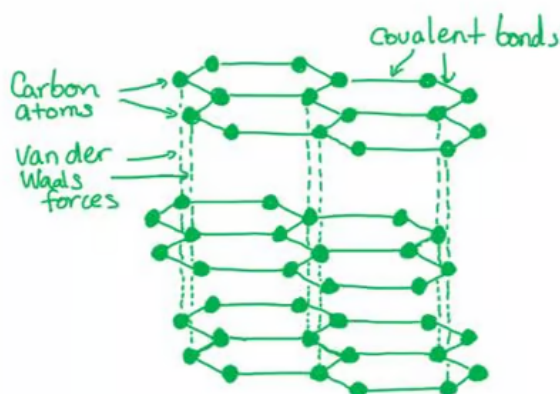
2

The correct answer is **B** because:

- Ionic compounds have the following properties:
 - Do not conduct electricity as a solid.
 - Will conduct electricity as a liquid or molten.
 - High melting and boiling points.
- Sodium chloride will have a lower melting and boiling point than magnesium oxide as sodium chloride is only bonding with one bonding pair of electrons whereas magnesium oxide is bonding with two bonding pair of electrons.



- Giant molecular structures have the following properties:
 - Solid at room temperature.
 - Very high melting and boiling points.
- Graphite will conduct electricity due to delocalised electrons between the layers.



A is incorrect as the formula for calcium fluoride is incorrect, it should be CaF_2 .

C & D are incorrect as Silicon dioxide and diamond do not conduct electricity, due to the strong covalent bonds in both structures.

3

The correct answer is **B** because:

- The ideal gas equation is $pV = nRT$.
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).

- **Rearranged for pressure:**

- $p = \frac{nRT}{V}$

- **Number of moles of hydrogen:**

- $\text{moles } (n) = \frac{\text{mass } (m)}{\text{molar mass } (M_r)} = \frac{0.96}{2} = 0.48$

- **Known quantities:**

- Number of moles; 0.48
 - Volume; $7.0 \times 10^{-3} \text{ m}^3$
 - Temperature; 303 K

- **Substitution into the equation:**

- $p = \frac{nRT}{V} = \frac{0.48 \times 8.314 \times 303}{7.0 \times 10^{-3}} = 172741.2 \text{ Pa} = 172.7 \text{ kPa}$

4

The correct answer is **B** because:

- **The ideal gas equation is $pV = nRT$.**

- Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).

- **Rearranged for number of moles:**

- $n = \frac{pV}{RT}$

- **Known quantities:**

- Pressure; 103000 Pa
 - Volume; $5730 \text{ cm}^3 = 0.00573 \text{ m}^3$
 - Temperature; $60^\circ\text{C} = 333 \text{ K}$

- **Substitution into the equation:**

- $n = \frac{103000 \times 0.00573}{8.314 \times 333} = 0.19978$

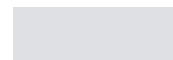
- **Mass of oxygen (O_2 $M_r = 32$):**

- $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.19978 \times 32 = 6.4 \text{ g}$

5

The correct answer is **D** because:

- The ideal gas equation is $pV = nRT$.



- **Known quantities:**

- Pressure; $1.00 \times 10^5 \text{ Pa}$
- Volume; $1.247 \times 10^{-3} \text{ m}^3$
- $\text{moles } (n) = \frac{\text{mass } (m)}{\text{molar mass } (M_r)} = \frac{5.35}{71} = 0.075$

- **Substitution into the equation:**

- $T = \frac{(1.0 \times 10^5) \times (1.247 \times 10^{-3})}{0.075 \times 8.314} = 199.98 \text{ K}$

6

The correct answer is **A** because:

- The temperature and number of moles remain constant throughout.
- **Boyles Law:** $P_1V_1 = P_2V_2$
 - Where: P_1 is the initial pressure, V_1 is the initial volume, P_2 is the final pressure, V_2 is the final volume
- **Therefore:**

- $V_2 = \frac{P_1}{P_2} V_1$

- **Known quantities:**

- Pressure (P_1); 101000 Pa
- Pressure (P_2); 2020000 Pa
- Volume (V_1); $200 \text{ cm}^3 = 0.0002 \text{ m}^3$

- **Substitution into the equation:**

- $V_2 = \frac{101000}{2020000} \times 0.0002 = 0.00001 \text{ m}^3 = 10 \text{ cm}^3$

7

The correct answer is **B** because:

- **The ideal gas equation is $pV = nRT$.**

- Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).
- **Rearranged for number of moles:**
 - $n = \frac{pV}{RT}$
- **Known quantities:**
 - Pressure; 100000 Pa
 - Volume; $960 \text{ cm}^3 = 0.00096 \text{ m}^3$
 - Temperature; $20^\circ\text{C} = 293 \text{ K}$
- **Substitution into the equation:**
 - $n = \frac{100000 \times 0.00096}{8.314 \times 293} = 0.0394$
- 2 moles of sodium produce 1 mole of H_2 gas therefore 0.0788 moles of sodium was used.
- **Mass of Sodium ($\text{Na } M_r = 23$):**
 - $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.0775 \times 23 = 1.81 \text{ g}$

8

The correct answer is **A** because:

- A temperature of 298 K will be room temperature; 25°C .
- A pressure of 101 kPa is 1 atmosphere.
- Halogenoalkanes have weakly polar bonds due to the differences in electronegativities; this creates a weak but permanent dipole.
- The presence of only weak forces means lower boiling point and gas at 298 K .

B is incorrect as amines have an N-H bond which will allow hydrogen bonding, this is a stronger bond, meaning that 1-aminopropane will be a liquid at 298 K .

C is incorrect as alcohols are permanently polarised due to the O-H bond and will have hydrogen bonds between the molecules, meaning that propan-1-ol will be a liquid at 298 K .

D is incorrect as diols have two O-H bonds in their structures and will have hydrogen bond between the molecules, meaning that ethane-1, 2-diol will be a liquid at 298 K.

9

The correct answer is **B** because:

- Ideal gas behaviour is seen with identical particles with no internal structure, no volume, random motion, no intermolecular forces and no loss of momentum in collisions.
- As the pressure increases the volume decreases, showing the pressure of the gas is **inversely proportional** demonstrated by **Boyle's Law**:
 - $P_1 V_1 = P_2 V_2$
 - Where: P_1 is the initial pressure, V_1 is the initial volume, P_2 is the final pressure, V_2 is the final volume.
- As the pressure increases the chances of inelastic collisions between the particles increases, so the gas behaves non-ideally.

A is incorrect as the increase in pressure is not going to make the real gas behave more like an ideal gas.

C is incorrect as the particles will not be absorbed by the vessel wall, but they are more likely to collide inelastically.

D is incorrect as the gas is behaving less like an ideal gas if it is partially liquified.

10

The correct answer is **D** because:

- **The ideal gas equation is $pV = nRT$.**
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).
- **Rearranged for number of moles:**
 - $n = \frac{pV}{RT}$
- **Known quantities:**

- Pressure; 1 atmospheres = 101000 Pa
- Volume; $3 \text{ dm}^3 = 0.003 \text{ m}^3$
- Temperature; $25^\circ\text{C} = 298 \text{ K}$
- **Substitution into the equation:**

$$101000 \times 0.003$$

likely to collide inelastically.

D is incorrect as the gas is behaving less like an ideal gas if it is partially liquified.

10

The correct answer is **D** because:

- **The ideal gas equation is $pV = nRT$.**
 - Where p is pressure (Pa), V is the volume (m^3), n is the number of moles, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), and T is the temperature (K).
- **Rearranged for number of moles:**
 - $n = \frac{pV}{RT}$
- **Known quantities:**
 - Pressure; 1 atmospheres = 101000 Pa
 - Volume; $3 \text{ dm}^3 = 0.003 \text{ m}^3$
 - Temperature; $25^\circ\text{C} = 298 \text{ K}$
- **Substitution into the equation:**
 - $n = \frac{101000 \times 0.003}{8.314 \times 298} = 0.12$
- **Mass of methane gas:**
 - $\text{mass } (m) = \text{moles } (n) \times \text{Molar mass } (M_r) = 0.12 \times 16 = 1.92 \text{ g}$
 - So the closest option is **D**

A is incorrect as mass of nitrogen gas: $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.12 \times 28 = 3.36 \text{ g}$

B is incorrect as mass of carbon dioxide gas: $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.12 \times 44 = 5.28 \text{ g}$

C is incorrect as mass of helium gas: $\text{mass } (m) = \text{moles } (n) \times \text{molar mass } (M_r) = 0.12 \times 4 = 0.48 \text{ g}$

Exam tip: make sure the units are converted; pressure, Pa, volume, m^3 , temperature K

11

The correct answer is **B** because:

- Ideal gas behaviour is seen with identical particles with no internal structure, no volume, random motion, no intermolecular forces and no loss of momentum in collisions.
- The ideal gas will have a straight line as the volume for an ideal gas is negligible.
- Real gases have intermolecular forces and the larger the forces, the less like an ideal gas.
- Hydrogen does have intermolecular forces, but they are weak, and the size of the molecule is small, so this is most likely to be X.
- Nitrogen is a diatomic molecule with weak dispersion forces between the particles, but it is a larger molecule than hydrogen so will deviate from ideal gas behaviour to a larger extent, so likely to be W.
- Ammonia has hydrogen bonding between the molecules, which is a stronger intermolecular force.
- Ammonia is going to deviate more from ideal gas behaviour so will be line Y.

12

The correct answer is **C** because:

- **The ideal gas equation is $pV = nRT$.**
 - Where p is pressure (Pa), V is volume (m^3), n is the number of moles, R is the gas constant ($8.314 J K^{-1} mol^{-1}$) and T is temperature (K).
- **Rearranged for pressure:**
 - $p = \frac{nRT}{V}$
- **Known quantities:**
 - Volume; $2 dm^3 = 0.002 m^3$
 - Temperature; $275 ^\circ C + 273 = 548 K$
 - Number of moles (n) = $1.5 \times 10^{-2} mol$
- **Substitution into the equation:**

$$\circ \quad p = \frac{nRT}{V} = \frac{1.5 \times 10^{-2} \times 8.31 \times (275 + 273)}{2 \times 10^{-3}}$$

Exam tip: make sure the units are converted; pressure, Pa, volume, m³, temperature K.

